Topic Modeling, Sentiment and Election Markets

Applied LSA and sentiment analysis around the 2026 US midterms

FRE-GY 7871 · Assignment 5 · sample 2026-03-02 to 2026-10-09 (155 NYSE sessions) · 5,816 filtered tweets · built 2026-10-10T12:34

Latest consensus P(Dem): House 90.3%, Senate 61.8%, Democratic sweep 61.0%.

Executive summary

Stage 1 (sentiment → election odds). 0 of 21 tests survive the Benjamini–Hochberg correction at 10% (smallest q = 0.56); 1 of 21 reject at an unadjusted 5% level, about what chance alone delivers. At a daily horizon, public issue sentiment on X does not measurably lead prediction-market prices.

Covariance SVD. No factor of the lagged sentiment–return cross-covariance beats its circular-shift permutation null (p = 0.73, 0.24, 0.54); the leading factor's out-of-sample correlation is -0.29 (t = -2.3), the opposite sign to its training fit, which is what over-fitting looks like.

News check (GDELT). 0 of 18 tests of per-issue news salience and news tone on ΔProb survive FDR at 10% (smallest q = 0.73). X discourse tracks the news agenda most closely for Healthcare (ρ = 0.49), Healthcare Affordability (ρ = 0.15), Energy/Climate (ρ = -0.05); the median level correlation across issues is 0.15. Series not available from GDELT: Economy/Inflation, Housing Affordability, Immigration, Foreign Policy/Defense, Technology/AI.

Stage 2 (election odds → assets). 7 of 57 excess-return tests and 0 of 57 volatility tests survive FDR at 10%.

Robustness. 5 of 7 survivors pass every check (HAC errors; without the 5 largest ΔProb sessions; without the 5 Jun 2026 sell-off; same sign in both halves; null placebo): ΔP(Sweep) → XLF (Σγ = -0.22, p = 0.003, both halves); ΔP(Sweep) → XLK (Σγ = +0.29, p = 0.002, mainly the second half); ΔP(Sweep) → UNH (Σγ = -0.46, p = 0.002, both halves); ΔP(Sweep) → NVDA (Σγ = +0.38, p = 0.006, mainly the second half); ΔP(Senate) → META (Σγ = +0.53, p = 0.004, mainly the second half). A rise in P(Dem senate) is followed the next session by outperformance of META. A rise in P(Dem sweep) is followed the next session by underperformance of XLF, UNH and outperformance of XLK, NVDA. The other survivors rest on a few extreme sessions or one half of the sample.

Rigobon–Sack. Relevance test p-values (Var_H/Var_L of ΔProb): House 0.629, Senate 0.034, Sweep 0.006. Even where the variance shift is significant it is too small to pin β down: 0 of 57 estimates have a 95% CI excluding zero, so contemporaneous causal effects are not identified in this sample.

Granger tests are predictive restrictions on lagged information, not structural effects. With one election cycle, read the q-values and the robustness battery rather than single p-values.

Data and provenance

Prediction-market coverage (sessions with a quote at or before 16:00 ET)

venue	House	Senate	Sweep
kalshi	155	155	155
polymarket	155	155	155
predictit	64	64	0

Tweet quality-filter waterfall

step	remaining	removed
raw	5,882	0
duplicate ids	5,882	0
non-English	5,882	0
retweets	5,882	0
followers < 10	5,882	0

step	remaining	removed
views < 100	5,816	66
astroturf (same text, >= 5 accounts)	5,816	0
fewer than 4 tokens	5,816	0

Missing prints after alignment to the NYSE calendar: AAPL 0, ENPH 0, GOOGL 0, ICLN 0, ITA 0, ITB 0, JPM 0, LMT 0, META 0, MSFT 0, NVDA 0, SPY 0, UNH 0, XAR 0, XHB 0, XLE 0, XLF 0, XLK 0, XLV 0, XOM 0, ^VIX 0, DGS2 1, DGS3MO 1. Vendor triangulation, staleness and outage checks are in DATA_AUDIT.md.

Figure 1. Implied probability of Democratic control, 16:00 ET quotes

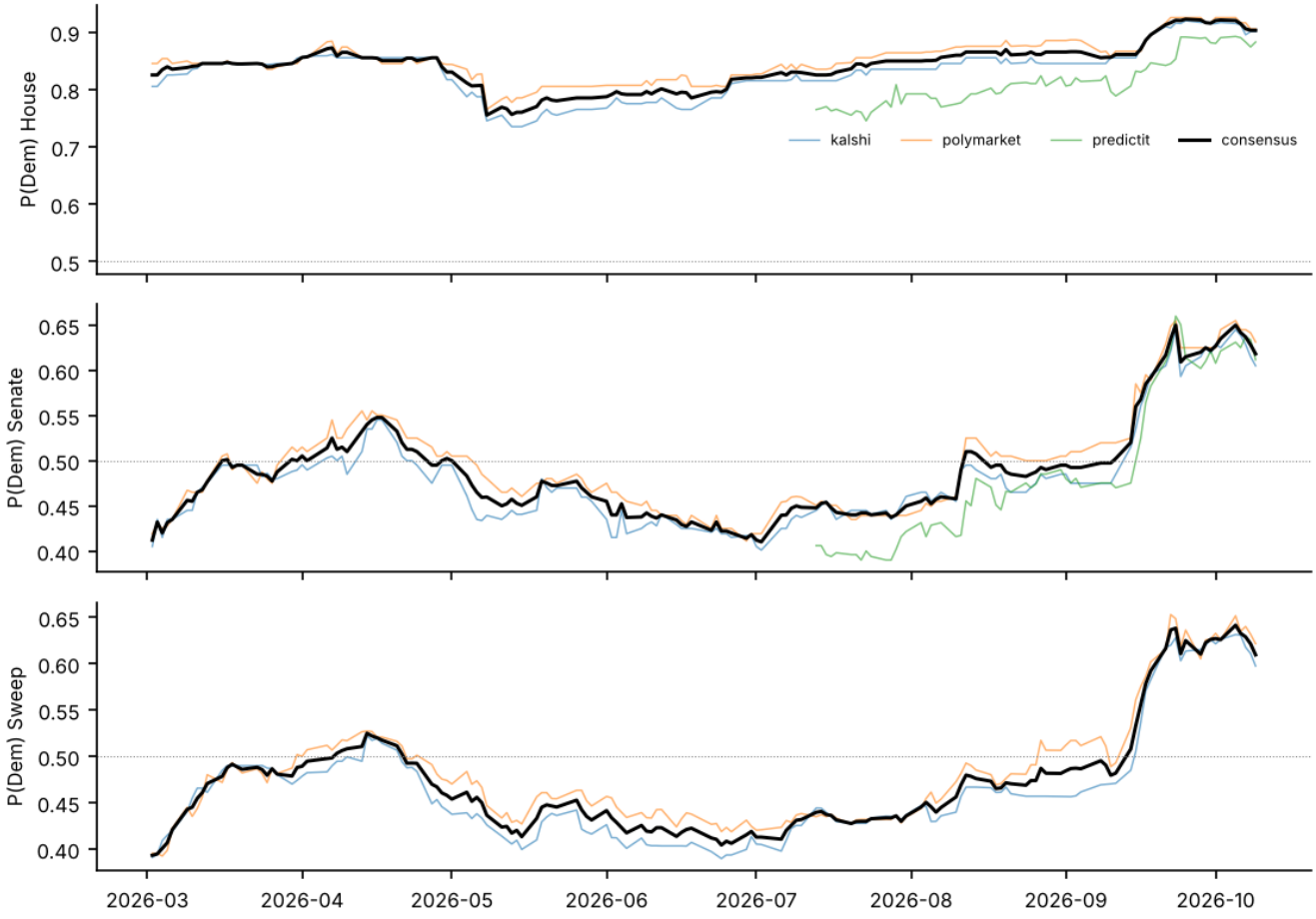
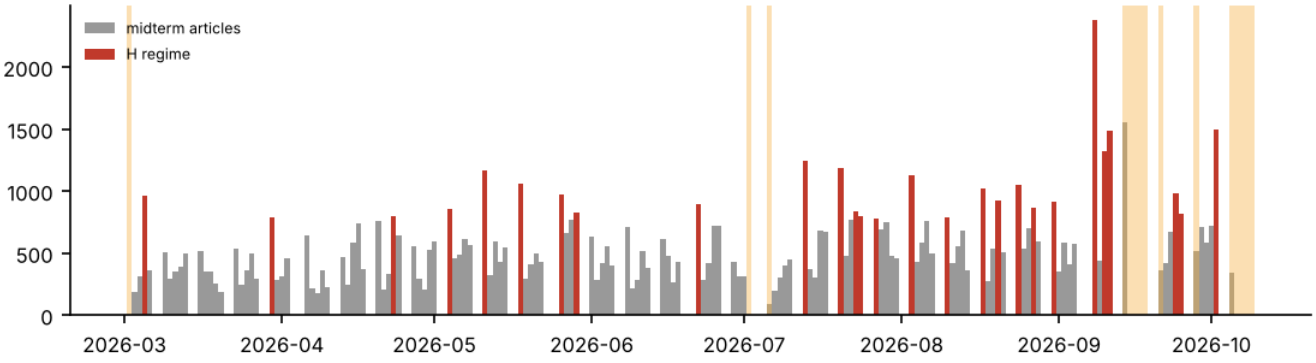


Figure 8. GDELT midterm coverage per session: H regime (red), GDELT outages (orange, excluded)



Topics

Table 1A. SVD latent topics and term loadings (U_k)

Issue	Axis	σ	Seed share	Purity	Stability	Top terms (loading)
Economy/Inflation	23+	3.07	0.71	0.97	1.00	tariffs (0.73); trump's (0.17); trump's tariffs (0.13); trade (0.13); trump tariffs (0.12); canada (0.11)
Economy/Inflation	6+	4.06	0.74	0.93	1.00	gas (0.48); prices (0.39); gas prices (0.29); grocery (0.19); groceries (0.17); grocery prices (0.15)
Energy/Climate	21+	3.11	0.48	0.86	0.60	energy (0.50); climate (0.15); ballot (0.14); bills (0.13); costs (0.11); energy costs (0.10)
Foreign Policy/Defense	3+	4.63	0.79	0.94	1.00	war (0.52); iran (0.46); war iran (0.24); iran war (0.18); military (0.17); billion (0.09)
General	0+	9.79	0.07	0.36	–	trump (0.19); republicans (0.19); health (0.16); democrats (0.15); congress (0.13); people (0.13)
Healthcare	10+	3.65	0.79	0.97	1.00	care (0.59); health care (0.53); health (0.30); affordable (0.11); care costs (0.09); affordable care (0.08)
Healthcare	10-	3.65	0.23	0.71	1.00	healthcare (0.12); insurance (0.09); health insurance (0.08); house (0.07); america (0.05); housing (0.03)
Healthcare	20+	3.13	0.31	0.90	1.00	medicare (0.43); social security (0.37); security (0.37); social (0.35); security medicare (0.20); medicare medicaid (0.14)
Healthcare	5+	4.09	0.72	0.99	1.00	health insurance (0.56); insurance (0.55); health (0.45); insurance premiums (0.08); insurance companies (0.07); insurance costs (0.06)
Healthcare	8+	3.77	0.41	0.94	1.00	medicaid (0.48); cuts (0.39); cut (0.18); medicaid cuts (0.16); big (0.16); tax cuts (0.15)
Healthcare	9+	3.71	0.55	0.71	1.00	healthcare (0.55); premiums (0.30); obamacare (0.20); costs (0.19); healthcare costs (0.14); medicare (0.12)
Healthcare Affordability	12+	3.50	0.25	0.73	1.00	obamacare (0.40); money (0.29); insurance companies (0.15); companies (0.14); pay (0.14); dollars (0.13)
Healthcare Affordability	26+	2.98	0.66	0.93	1.00	aca (0.46); subsidies (0.41); aca subsidies (0.31); expire (0.17); coverage (0.15); let (0.14)
Healthcare Affordability	7+	3.88	0.68	0.65	1.00	drug (0.48); drug prices (0.42); prices (0.36); prescription (0.29); prescription drug (0.27); lower (0.16)
Housing Affordability	15+	3.38	0.50	0.93	1.00	rent (0.54); rent control (0.35); ballot (0.32); control (0.31); massachusetts (0.15); groceries (0.13)
Housing Affordability	2+	4.71	0.60	0.99	1.00	housing (0.54); housing costs (0.37); costs (0.36); lower (0.20); lower housing (0.20); bipartisan (0.12)
Immigration	17+	3.27	0.69	0.99	1.00	immigration (0.61); illegal (0.28); enforcement (0.27); immigration enforcement (0.22); illegal immigration (0.16); immigrants (0.11)
Immigration	4+	4.46	0.64	0.99	1.00	border (0.54); ice (0.31); border patrol (0.29); patrol (0.28); ice border (0.22); fund (0.14)

Varimax-rotated LSA, k = 30, uncentred TF-IDF. Loadings are entries of U_k. Stability: share of 5 refits on 80% subsamples in which the pole reappears (same label, cosine ≥ 0.6).

Figure 2. LSA spectrum and topic stability

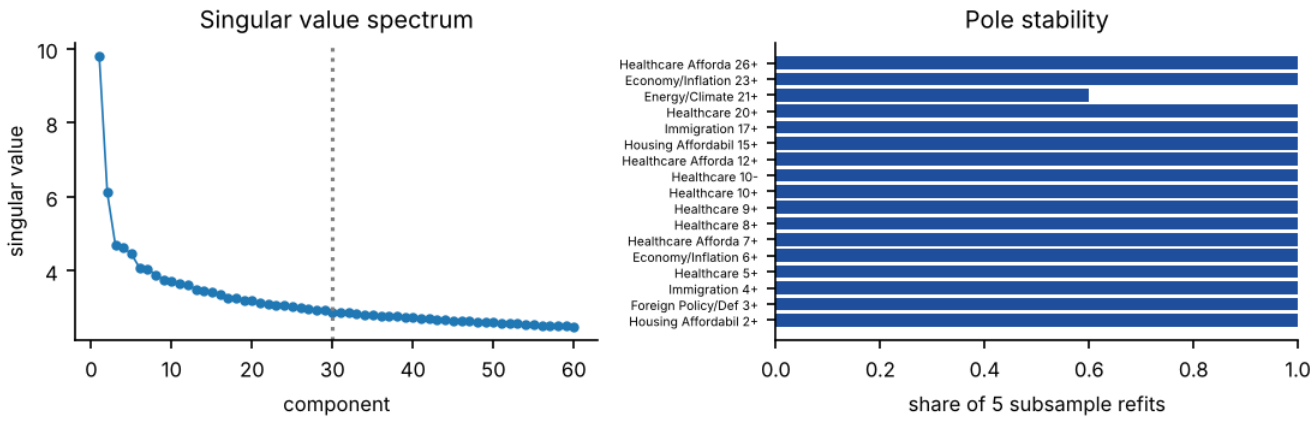


Figure 3. Term loadings U_k of the labelled poles (positive part)

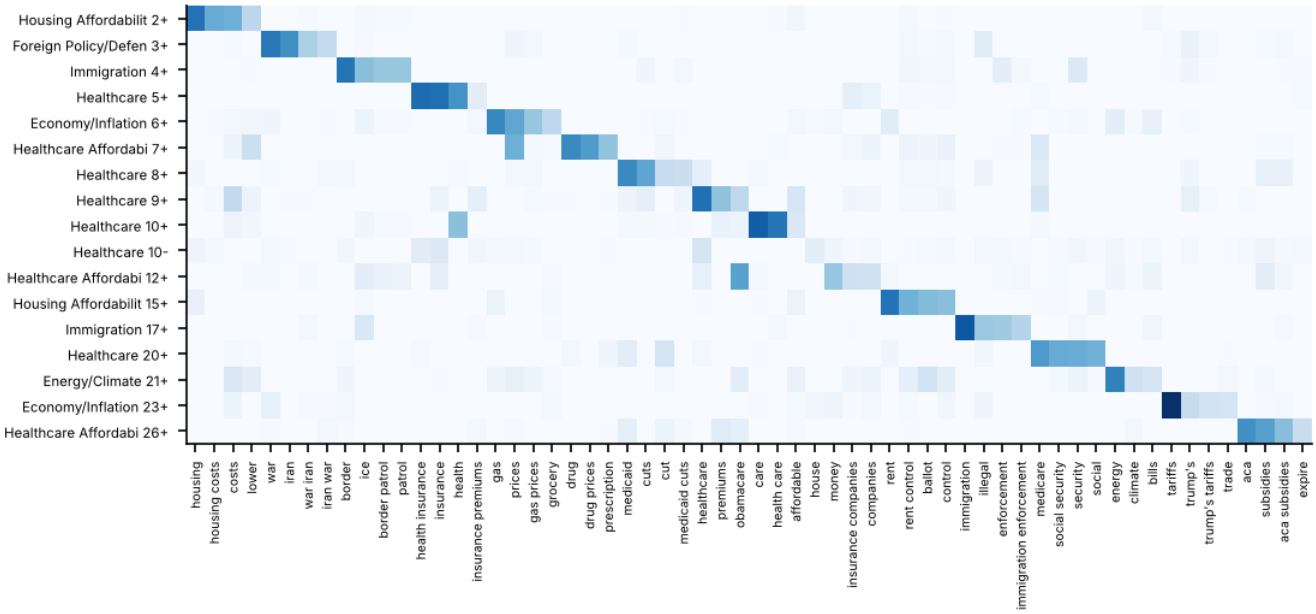
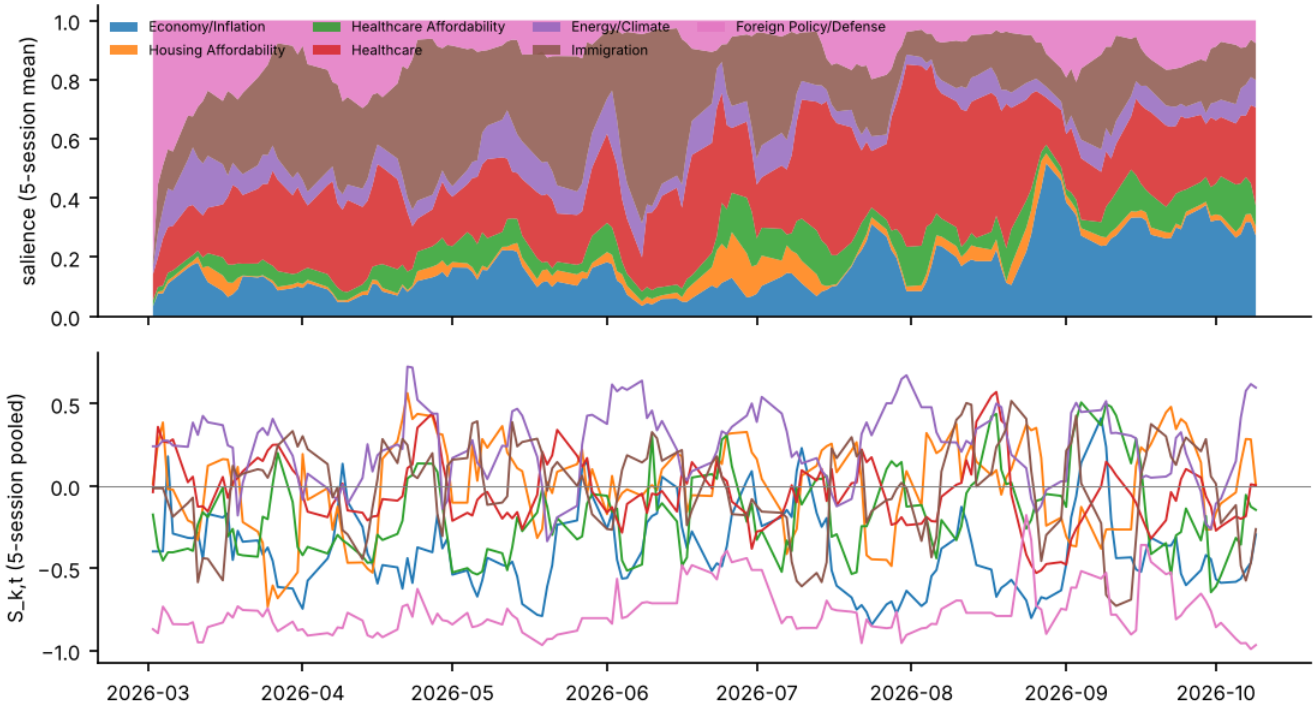


Table 1B. Topic salience and sentiment by issue

Issue	Docs	Salience	S mean	S sd	% Pos	% Neg	% Anxiety	% Anger	% Optimism	% Re-signed	Coverage	Used
Economy/Inflation	416	0.17	-0.40	0.25	25.00	71.88	14.90	6.01	14.66	13.46	0.98	yes
Housing Affordability	447	0.03	0.01	0.27	47.65	42.06	13.87	3.36	15.88	9.40	0.98	yes
Healthcare Affordability	401	0.07	-0.17	0.26	41.90	50.37	6.48	4.24	11.97	12.97	0.99	yes
Healthcare	1,076	0.27	-0.04	0.21	45.35	51.39	11.43	6.88	11.15	7.62	0.99	yes
Energy/Climate	160	0.08	0.24	0.23	72.50	26.25	13.75	3.12	31.87	5.00	0.96	yes
Immigration	480	0.25	-0.01	0.29	44.79	46.88	9.17	7.50	13.12	1.67	0.98	yes
Foreign Policy/Defense	269	0.12	-0.77	0.15	8.92	89.59	25.28	8.18	21.93	0.37	0.88	yes
Technology/AI	0	0.00	–	–	–	–	–	–	–	–	0.00	no (too sparse)

Salience: mean daily share of issue loading, general stratum only. S: 5-session pooled topic-conditioned sentiment. % Re-signed: share of documents whose issue-conditioned polarity differs from generic VADER.

Figure 4. Issue salience (general stratum) and topic-conditioned sentiment



Stage 1: sentiment → prediction markets

Table 2. Granger causality tests (topic sentiment → ΔProb)

Issue	Contract	Lags	N	F	p	q (BH)	Sig.	Σγ	ΔR²	p (3d)	p (rev)
Economy/Inflation	House	1	153	0.685	0.409	0.741		-0.002	0.003	0.428	0.069
Economy/Inflation	Senate	1	153	0.050	0.823	0.983		-0.000	0.000	0.437	0.534
Economy/Inflation	Sweep	1	153	0.063	0.801	0.983		-0.001	0.000	0.407	0.154
Energy/Climate	House	1	153	0.006	0.940	0.983		-0.000	0.000	0.169	0.064
Energy/Climate	Senate	1	153	0.489	0.485	0.784		-0.002	0.002	0.202	0.352
Energy/Climate	Sweep	1	153	0.876	0.351	0.741		-0.002	0.003	0.334	0.242
Foreign Policy/Defense	House	1	153	3.802	0.053	0.557	*	0.008	0.028	0.265	0.989
Foreign Policy/Defense	Senate	1	153	2.496	0.116	0.625		0.008	0.014	0.957	0.765
Foreign Policy/Defense	Sweep	1	153	0.260	0.611	0.917		0.002	0.001	0.724	0.927
Healthcare	House	3	151	1.025	0.384	0.741		-0.001	0.011	0.762	0.481
Healthcare	Senate	1	153	2.234	0.137	0.625		-0.005	0.011	0.033	0.605
Healthcare	Sweep	3	151	2.832	0.041	0.557	**	-0.000	0.036	0.178	0.105
Healthcare Affordability	House	1	153	0.994	0.320	0.741		0.003	0.011	0.452	0.990
Healthcare Affordability	Senate	1	153	0.006	0.939	0.983		-0.000	0.000	0.883	0.233
Healthcare Affordability	Sweep	1	153	0.004	0.947	0.983		0.000	0.000	0.798	0.485
Housing Affordability	House	1	153	0.000	0.983	0.983		-0.000	0.000	0.772	0.716
Housing Affordability	Senate	1	153	1.508	0.221	0.741		-0.004	0.011	0.060	0.688
Housing Affordability	Sweep	1	153	0.763	0.384	0.741		-0.002	0.005	0.546	0.096
Immigration	House	1	153	2.107	0.149	0.625		0.002	0.007	0.157	0.664
Immigration	Senate	1	153	0.645	0.423	0.741		-0.003	0.006	0.443	0.308
Immigration	Sweep	1	153	0.158	0.691	0.968		-0.001	0.001	0.676	0.895

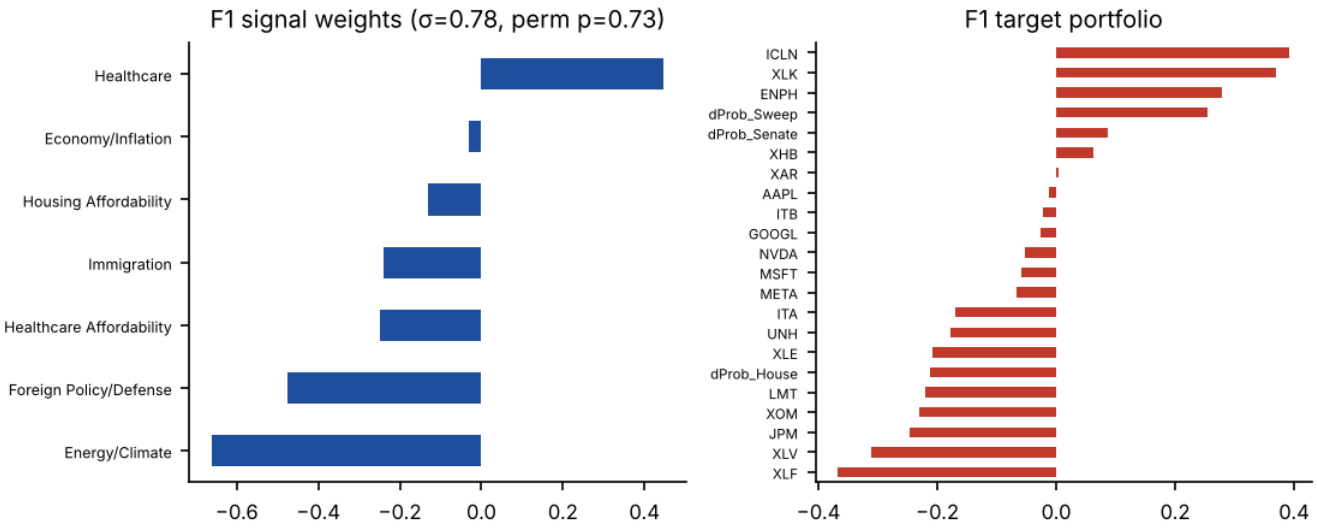
HC1 F-tests of lags of S_k in an AR model of ΔProb with lagged SPY, ΔVIX and Δ2Y controls; lags by BIC. q: BH FDR across the table. p (3d): 3-session pooled sentiment. p (rev): ΔProb → S_k. *** p<0.01 ** p<0.05 * p<0.10.

Covariance SVD (signal–return alpha factors)

index	σ	perm p	oos_days	oos_corr	oos_t	sign_hit_rate	targets_right_sign
F1	0.783	0.735	62	-0.285	-2.306	0.403	0.318
F2	0.702	0.240	62	0.031	0.240	0.484	0.727
F3	0.484	0.539	62	-0.076	-0.588	0.452	0.545

Estimated on sessions up to 2026-07-14; the out-of-sample statistics use the remaining sessions only.

Figure 6. Leading factor of the sentiment–return cross-covariance SVD



News salience check (GDELT)

Does X discourse track the news agenda? Correlation of X issue salience (general stratum) with GDELT's share of midterm coverage mentioning the issue, 5-session means.

issue	sessions	corr_levels	corr_5d_changes	x_mean_salience	news_mean_salience
Healthcare Affordability	146	0.147	0.179	0.071	0.022
Healthcare	146	0.493	-0.055	0.274	0.095
Energy/Climate	146	-0.053	-0.110	0.076	0.297

Table 2B. Granger causality tests (GDELT news salience and tone → ΔProb)

Issue	Series	Contract	Lags	N	F	p	q (BH)	Sig.	Σγ
Energy/Climate	salience	House	1	145	1.105	0.295	0.756		-0.005
Energy/Climate	salience	Senate	1	145	0.298	0.586	0.756		0.003
Energy/Climate	salience	Sweep	1	145	4.268	0.041	0.733	**	0.009
Energy/Climate	tone	House	1	145	0.329	0.567	0.756		0.000
Energy/Climate	tone	Senate	1	145	0.261	0.611	0.756		0.000
Energy/Climate	tone	Sweep	1	145	1.690	0.196	0.756		0.001
Healthcare	salience	House	1	145	1.773	0.185	0.756		0.012
Healthcare	salience	Senate	1	145	2.084	0.151	0.756		0.016
Healthcare	salience	Sweep	1	145	0.626	0.430	0.756		0.008
Healthcare	tone	House	1	145	0.155	0.694	0.756		-0.000
Healthcare	tone	Senate	1	145	0.001	0.981	0.981		0.000
Healthcare	tone	Sweep	1	145	0.135	0.714	0.756		0.000
Healthcare Affordability	salience	House	1	145	0.238	0.626	0.756		0.004
Healthcare Affordability	salience	Senate	1	145	0.209	0.648	0.756		0.010
Healthcare Affordability	salience	Sweep	1	145	0.214	0.644	0.756		-0.014
Healthcare Affordability	tone	House	1	139	0.541	0.463	0.756		-0.000
Healthcare Affordability	tone	Senate	1	139	0.149	0.700	0.756		-0.000
Healthcare Affordability	tone	Sweep	1	139	1.252	0.265	0.756		0.001

Same specification and FDR as Table 2. GDELT outage sessions are missing, not zero, and drop out. Availability of each GDELT series: Economy/Inflation (no counts, no tone), Housing Affordability (no counts, no tone), Healthcare Affordability (counts, tone), Healthcare (counts, tone), Energy/Climate (counts, tone), Immigration (no counts, no tone), Foreign Policy/Defense (no counts, no tone), Technology/AI (no counts, no tone).

Stage 2: prediction markets → midterm basket

Table 3. Midterm equity basket reactivity matrix

Asset	Contract	Ret p	Ret q	Ret Σγ	Vol p	Vol q	β OLS	β RS	RS 95% CI	VarH/VarL	Relevance p
XLE	House	0.428	0.735	-0.168	0.869	0.952	0.287	3.993	[-14.19, 10.85]	0.884	0.629
XLE	Senate	0.119	0.489	-0.275	0.040	0.446	0.093	-1.367	[-15.73, 13.05]	1.677	0.034
XLE	Sweep	0.319	0.667	-0.178	0.763	0.882	-0.099	0.092	[-5.61, 7.91]	2.017	0.006
ICLN	House	0.000	0.013	-0.477	0.027	0.385	-0.300	-3.062	[-12.40, 12.41]	0.884	0.629
ICLN	Senate	0.832	0.949	0.027	0.888	0.955	0.146	0.360	[-7.22, 6.80]	1.677	0.034
ICLN	Sweep	0.151	0.494	0.185	0.659	0.841	0.075	-0.146	[-8.28, 5.16]	2.017	0.006
XLV	House	0.170	0.509	0.162	0.514	0.841	0.197	2.423	[-8.90, 9.85]	0.884	0.629
XLV	Senate	0.802	0.949	-0.021	0.371	0.784	-0.053	-0.696	[-5.69, 4.44]	1.677	0.034
XLV	Sweep	0.016	0.115	-0.242	0.770	0.882	-0.147	-0.384	[-4.32, 3.24]	2.017	0.006
XLF	House	0.887	0.969	-0.014	0.418	0.802	0.181	-0.352	[-5.97, 4.93]	0.884	0.629
XLF	Senate	0.119	0.489	-0.097	0.583	0.841	-0.075	0.248	[-3.04, 3.72]	1.677	0.034
XLF	Sweep	0.003	0.045	-0.222	0.047	0.446	-0.187	0.142	[-2.30, 3.04]	2.017	0.006
XAR	House	0.464	0.735	-0.106	0.637	0.841	-0.017	2.573	[-10.22, 13.04]	0.884	0.629
XAR	Senate	0.717	0.949	0.039	0.655	0.841	0.078	-0.368	[-4.81, 4.73]	1.677	0.034
XAR	Sweep	0.937	0.981	0.009	0.149	0.564	0.111	-0.005	[-4.37, 4.94]	2.017	0.006
ITA	House	0.482	0.742	-0.080	0.723	0.877	0.043	1.269	[-5.52, 11.77]	0.884	0.629
ITA	Senate	0.837	0.949	-0.019	0.664	0.841	0.018	-0.049	[-3.75, 3.95]	1.677	0.034
ITA	Sweep	0.328	0.667	-0.098	0.246	0.668	0.030	0.168	[-4.22, 4.98]	2.017	0.006
XLK	House	0.353	0.673	-0.105	0.654	0.841	-0.196	-1.848	[-5.57, 6.57]	0.884	0.629
XLK	Senate	0.837	0.949	0.018	0.109	0.479	0.072	-0.285	[-5.52, 5.87]	1.677	0.034
XLK	Sweep	0.002	0.041	0.289	0.079	0.449	0.162	-0.167	[-4.66, 3.29]	2.017	0.006
ITB	House	0.778	0.949	0.056	0.540	0.841	0.141	-4.186	[-9.57, 17.64]	0.884	0.629
ITB	Senate	0.901	0.969	-0.015	0.954	0.975	-0.111	0.987	[-11.92, 15.13]	1.677	0.034
ITB	Sweep	0.947	0.981	-0.009	0.469	0.835	0.025	0.300	[-3.42, 6.76]	2.017	0.006
XHB	House	0.787	0.949	-0.046	0.344	0.762	0.121	-3.187	[-7.55, 14.22]	0.884	0.629
XHB	Senate	0.967	0.984	-0.005	0.717	0.877	-0.115	0.805	[-9.75, 14.34]	1.677	0.034
XHB	Sweep	0.849	0.949	-0.024	0.312	0.762	0.042	0.058	[-2.66, 4.40]	2.017	0.006
ENPH	House	0.145	0.494	-0.692	0.244	0.668	-0.645	-1.156	[-24.34, 23.61]	0.884	0.629
ENPH	Senate	0.740	0.949	-0.107	0.594	0.841	0.036	0.407	[-12.99, 15.66]	1.677	0.034
ENPH	Sweep	0.129	0.489	0.588	0.390	0.793	-0.328	-0.304	[-9.11, 8.90]	2.017	0.006
XOM	House	0.374	0.687	-0.223	0.652	0.841	0.432	5.266	[-18.92, 15.78]	0.884	0.629
XOM	Senate	0.125	0.489	-0.319	0.065	0.449	0.113	-1.348	[-14.76, 13.09]	1.677	0.034
XOM	Sweep	0.327	0.667	-0.203	0.629	0.841	-0.130	-0.287	[-7.42, 6.87]	2.017	0.006
UNH	House	0.272	0.621	0.340	0.958	0.975	-0.097	3.288	[-14.77, 9.57]	0.884	0.629
UNH	Senate	0.463	0.735	-0.122	0.328	0.762	-0.053	-0.907	[-7.78, 7.10]	1.677	0.034
UNH	Sweep	0.002	0.041	-0.456	0.087	0.453	-0.438	-0.402	[-3.74, 3.86]	2.017	0.006
JPM	House	0.683	0.949	-0.071	0.074	0.449	0.305	0.090	[-6.85, 5.86]	0.884	0.629
JPM	Senate	0.189	0.538	-0.126	0.594	0.841	-0.118	-0.027	[-3.10, 4.58]	1.677	0.034
JPM	Sweep	0.030	0.187	-0.267	0.010	0.385	-0.191	-0.211	[-4.00, 4.27]	2.017	0.006
LMT	House	0.545	0.796	-0.118	0.348	0.762	0.228	13.212	[-41.88, 43.12]	0.884	0.629
LMT	Senate	0.460	0.735	-0.106	0.996	0.996	0.086	-0.229	[-8.12, 6.13]	1.677	0.034
LMT	Sweep	0.413	0.735	-0.128	0.958	0.975	0.037	0.699	[-5.50, 9.87]	2.017	0.006
NVDA	House	0.998	0.998	0.000	0.158	0.564	0.095	1.438	[-8.43, 9.36]	0.884	0.629
NVDA	Senate	0.007	0.059	0.303	0.773	0.882	0.099	-0.062	[-5.68, 4.49]	1.677	0.034

Asset	Contract	Ret p	Ret q	Ret Σγ	Vol p	Vol q	β OLS	β RS	RS 95% CI	VarH/VarL	Relevance p
NVDA	Sweep	0.006	0.058	0.379	0.106	0.479	0.414	-0.129	[-10.07, 6.96]	2.017	0.006
MSFT	House	0.198	0.538	0.231	0.442	0.813	0.278	4.005	[-13.31, 11.42]	0.884	0.629
MSFT	Senate	0.237	0.588	-0.161	0.208	0.668	0.241	0.783	[-7.35, 10.87]	1.677	0.034
MSFT	Sweep	0.156	0.494	-0.219	0.138	0.562	-0.070	1.459	[-11.71, 18.30]	2.017	0.006
GOOGL	House	0.644	0.918	0.139	0.796	0.890	-0.391	-1.889	[-14.65, 11.25]	0.884	0.629
GOOGL	Senate	0.232	0.588	0.115	0.422	0.802	0.072	0.343	[-10.15, 11.84]	1.677	0.034
GOOGL	Sweep	0.265	0.621	-0.160	0.291	0.754	0.081	0.752	[-10.58, 15.64]	2.017	0.006
META	House	0.047	0.267	0.631	0.068	0.449	0.043	4.428	[-14.51, 11.10]	0.884	0.629
META	Senate	0.004	0.051	0.534	0.022	0.385	-0.151	-0.844	[-12.85, 10.15]	1.677	0.034
META	Sweep	0.069	0.359	0.408	0.213	0.668	-0.181	-0.727	[-8.62, 7.20]	2.017	0.006
AAPL	House	0.502	0.754	-0.100	0.553	0.841	-0.059	-0.646	[-5.71, 8.53]	0.884	0.629
AAPL	Senate	0.354	0.673	-0.101	0.022	0.385	-0.226	0.469	[-4.24, 7.54]	1.677	0.034
AAPL	Sweep	0.792	0.949	0.040	0.246	0.668	-0.490	-0.071	[-6.31, 4.25]	2.017	0.006

Ret: excess return over SPY. Vol: 2(H−L)/(H+L), differenced where ADF does not reject a unit root. q: BH across each 57-test family. β RS: Rigobon–Sack estimate; VarH/VarL and its p-value are the relevance condition.

Robustness battery for every FDR survivor

stage	cause	effect	sum_gamma	p	p_HAC	p_no_extremes	p_excl_dates	p_1st_half	p_2nd_half	p_placebo	robust
2 ret	dProb_House	ICLN	-0.477	0.000	0.000	0.149	0.000	0.001	0.461	0.977	False
2 ret	dProb_Sweep	XLF	-0.222	0.003	0.002	0.040	0.005	0.062	0.039	0.785	True
2 ret	dProb_Sweep	XLK	0.289	0.002	0.002	0.006	0.003	0.129	0.003	0.190	True
2 ret	dProb_Sweep	UNH	-0.456	0.002	0.002	0.004	0.003	0.015	0.064	0.465	True
2 ret	dProb_Senate	NVDA	0.303	0.007	0.008	0.124	0.002	0.763	0.002	0.290	False
2 ret	dProb_Sweep	NVDA	0.379	0.006	0.006	0.051	0.009	0.405	0.003	0.185	True
2 ret	dProb_Senate	META	0.534	0.004	0.003	0.051	0.003	0.383	0.019	0.952	True

Figure 5. Granger tests: FDR q-values (darker = smaller)

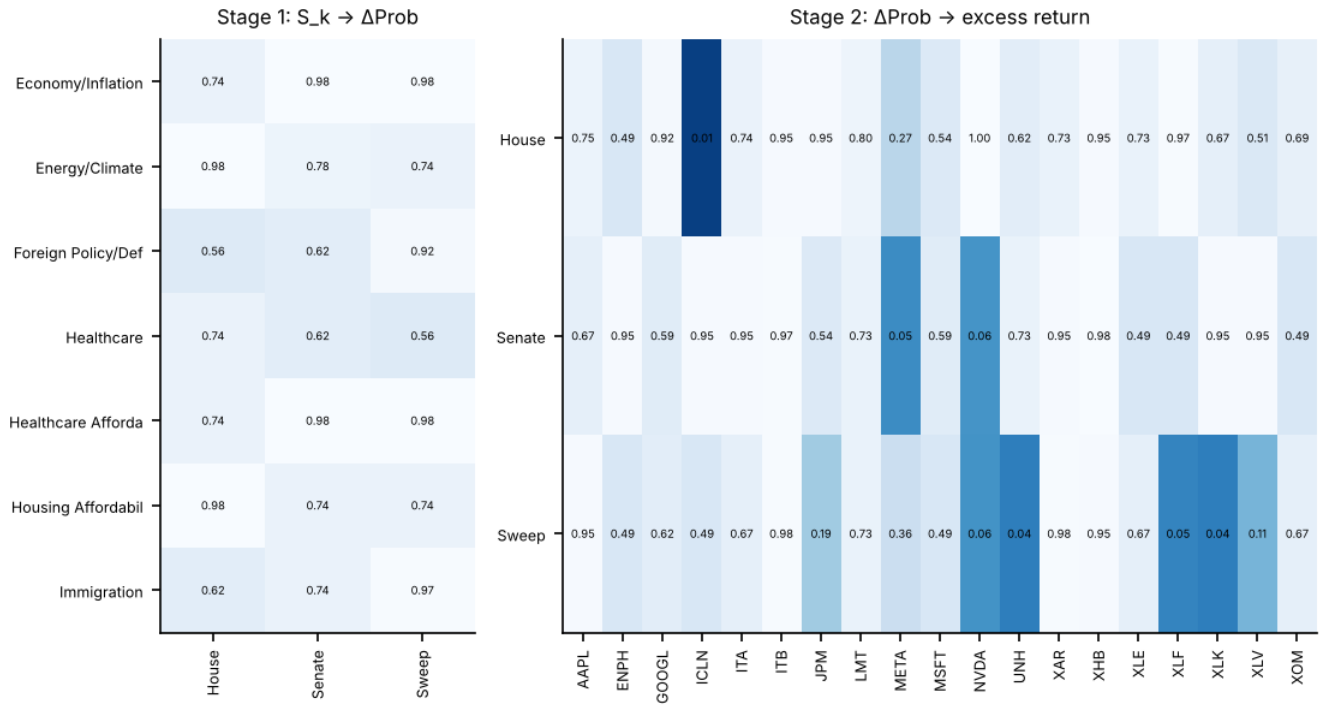
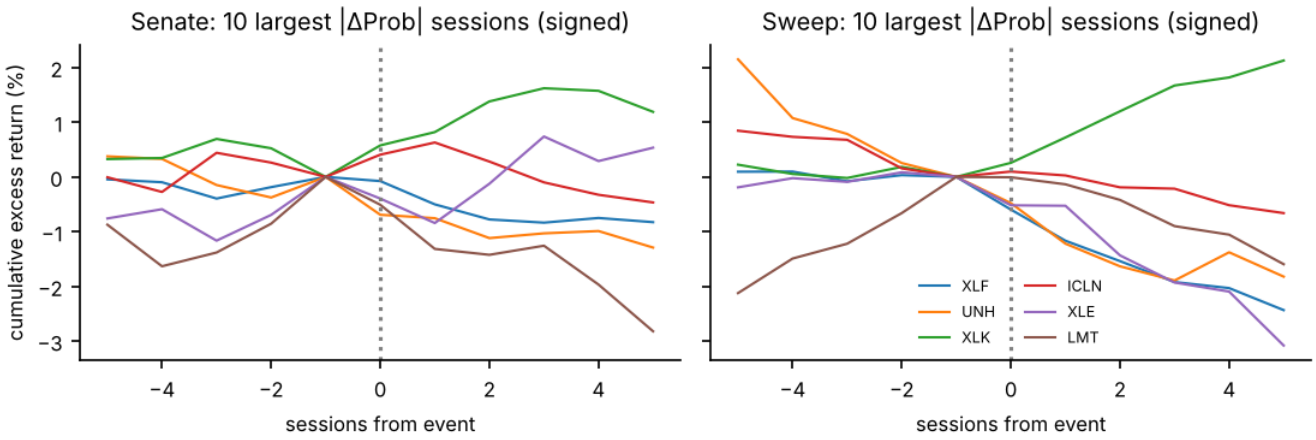


Figure 7. Basket reactions around large prediction-market moves



Methods

Calendar and timing. Everything lives on the NYSE calendar. Tweets are fetched per session (previous 16:00 ET close to this one), so weekend text lands on Monday. Prediction-market probabilities are the last hourly quote at or before 16:00 ET (median staleness: Kalshi 0 h, Polymarket 1 h). A GDELT day ends at 20:00 ET and counts for the next session. Returns never span a missing print.

Prediction markets. Kalshi (CONTROLH-2026, CONTROLS-2026, KXBALANCEPOWERCOMBO-27FEB) and Polymarket (control and balance-of-power markets), each normalised across all of its outcomes. Consensus = equal-weight average; PredictIt (daily bars only) is a cross-check with zero weight.

Text. twitterapi.io advanced search, two strata per session: a general issue × political-context query, and an affordability oversample (housing and healthcare costs) used for tone only. Server-side: English, no retweets or replies, 5–150 likes. Client-side: followers ≥ 10, views ≥ 100, astroturf (same text from ≥ 5 accounts). Budget: 93,450 credits, every call ledgered.

LSA. Uncentred TF-IDF (unigrams, bigrams, hashtags, cashtags; sublinear tf) → truncated SVD, k = 30 → varimax rotation of components 1..29. Centring would densify the matrix and treat unseen terms as negative evidence. Each pole is labelled by the share of its loading energy on an issue's seed terms (≥ 15%, purity ≥ 60%); stability = reappearance across 5 refits on 80% subsamples.

Sentiment. A document's weight on issue k is the squared cosine of its projection on k's poles. Polarity is VADER with an issue-specific overlay (e.g. 'rising' is negative for prices and rents). $S_{k,t} = \Sigma(\text{Pos} - \text{Neg})/\Sigma\text{Total}$ pooled over 5 sessions with ≥ 5 effective documents; gaps are filled only from the past.

Econometrics. ADF-tested series; Granger F-tests with HC1 errors, BIC lags and lagged SPY/ΔVIX/Δ2Y controls; Benjamini–Hochberg FDR per table; a robustness battery for every survivor. Covariance SVD of $(1/T)\Sigma\, s_{t-1}'\delta y_t$ on the first 60% of the sample, circular-shift permutation test, out-of-sample scoring on the remaining 40%. Rigobon–Sack $\beta = \Delta\text{Cov}/\Delta\text{Var}$ across H (top 20% GDELT midterm-news sessions) and L sessions, GDELT outage sessions excluded, regime bootstrap CI.

Topic-model comparison (baselines)

method	components	umass_coherence	topic_diversity	share_labelled	issues_found
NMF	12	-1.772	0.892	0.692	6
LDA	12	-1.724	0.658	0.686	5

Limitations

One campaign: 155 sessions. A daily Granger test has little power against effects that build over weeks.

The X sample is politically engaged mid-size accounts (median ~24–42k followers), not the median voter: posts that earn ≥ 5 likes mostly come from accounts with an audience.

Technology/AI never formed a topic, so the Big Tech basket is only tested against election odds.

GDELT's feed had 19 outage days (mid-September and most of October); those sessions are excluded from the regime split rather than read as quiet news days.

Yahoo daily closes for the S&P; 500 trackers disagree by 15–30bp a day over 25 Jun – 1 Jul 2026 (they agree on the week's total). Using the median of SPY/IVV/VOO as the benchmark, or dropping that week, changes no Stage 2 conclusion.

The robust Stage 2 results are lagged predictability. They fit slow pricing of regulatory risk, but with this many tests and one cycle they are hypotheses to test in 2028, not a trading rule.

AI use

Built with Claude Code. What was done by the model and what by the author is disclosed in AI_USE.md.